NANYANG TECHNOLOGICAL UNIVERSITY

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***U**ndergraduate **R**esearch **E**xperience on **CA**mpus*

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Numerical Study of Conductance Properties of 2D Topological Insulators

Background and Motivation

A solid with the properties of being an insulator on the inside and a conductor on the outside surfaces were first observed in 1980 as the **quantum Hall effect** in a presence of a strong magnetic field [1]. What makes these properties interesting is the robustness of the conducting surface; they are topologically protected, deformities or impurities on the surface can not stop or fluctuate the flowing current.

A **topological insulator** is a material that has properties similar to the quantum Hall effect without the presence of a magnetic field (ie. unbroken time-reversal symmetry). This is achieved when the electron spin can 'replace' the magnetic field. Physicists have been looking for these types of materials for the past few decades with the first 2D topological insulator discovered in 2007 [2].

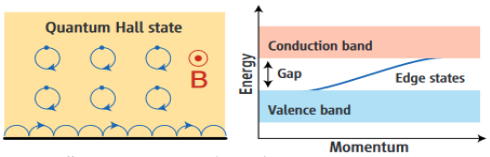

Fig. 1: Quantum Hall effect. Electrons at the edge 'bounces' and travel to the right, while electrons in the centre bulk are 'locked' as they revolves in place. Image courtesy of Charles L. Kane and Eugene J. Mele. [3]

Topological protection ensures non-fluctuating currents and robustness against surface impurities and deformities; both are useful features to have. Although practical applications are mostly speculative, these unique properties might prove to be promising in the fields of electronic, spintronic and quantum computing [4][5].

Methodology

In this research project, the properties of different lattice structures will be investigated. A lattice will have its solutions derived both analytically and numerically. Both results can be compared and visualised. Through the process, properties of the lattice structure like energy spectrum, edge states and conductance can be revealed. It is then easy to identify properties that are different between conventional and topological insulators.

Using the **tight-binding method**, developed by Slater and Koster in 1954 [6], the wave functions of lattices can be easily approximated analytically along with its energy eigenvalue. Shown below is the general tight-binding Hamiltonian for a honeycomb lattice and its diagonalised expression derived by Haldane in 1988 [7] and is thus known as the Haldane model.

$$H = t_1 \sum_{ij} c_i^+ c_j + t_2 \sum_{ij} e^{-iv_{ij}\phi} c_i^+ c_j + M \sum_{ij} \epsilon_i c_i^+ c_j$$

$$\boldsymbol{H}(\boldsymbol{k}) = 2t_2 cos\phi \left[\sum_i \cos(\boldsymbol{k}.\boldsymbol{b}_i)\right] \boldsymbol{I} + t_1 \left[\sum_i [\cos(\boldsymbol{k}.\boldsymbol{a}_i)\boldsymbol{\sigma}_x + \sin(\boldsymbol{k}.\boldsymbol{a}_i)\boldsymbol{\sigma}_y]\right] + \left[M - 2t_2 sin\phi \left[\sum_i \sin(\boldsymbol{k}.\boldsymbol{b}_i)\right]\right] \boldsymbol{\sigma}_z$$

The lattices that are studied analytically is then computed and visualised on computer in Python using ***kwant*** package. *Kwant* is capable of plotting lattices of different types and also solving the wave function and energy eigenvalues of the lattice numerically. This allows the conductance, energy bands and edge states of a given lattice to be visualised.

Kwant comes with a library of solvers. The lattice must be built and finalised before a solver can be called. As an example, to compute the conductance, only five lines of code are needed:

```
energies=[0.01 * i for i in range(300)]
data = []
for energy in energies:
    smatrix = kwant.smatrix(sys, energy)
    data.append(smatrix.transmission(1, 0))
```

In this case, the lattice is built in 'sys'. The graph of conductance against energy can then be easily plotted with any graphing modules in Python by plotting 'data' against 'energies'.

Findings

One of the 2D lattices studied is the **Haldane model** mentioned above. The lattice has 2 unique edges; the armchair edge and the zig-zag edge. Figure 2 depicts the edges with the zig-zag edges at the top and bottom, and the armchair edges at the left and right of the lattice. From figure 2, it can be seen that the honeycomb lattice exhibits the quantum Hall effect as electrons has the highest probability of being found at the zig-zag edges.

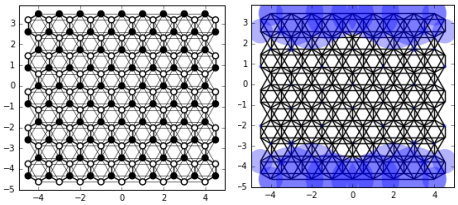

Fig 2: Honeycomb lattice and its electron probability distribution(blue) at the 98th eigen mode. Nearest-neighbours hopping is depicted with bolder lines than next-nearest-neighbours hopping. Plotted in Python using *Kwant*.

Figure 3 shows the conductance as a function of energy eigenvalues of the honeycomb lattice. It can be seen that the conductance takes on discrete values, another property similar to the quantum Hall effect. This is because the electrons are quantized and only travel with discrete radiuses.

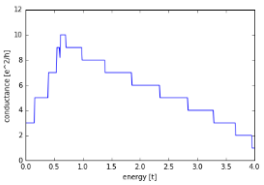

Fig 3: Conductance against energy graph of honeycomb lattice.

The numerical results show that the Haldane model exhibits properties similar to topological insulators with edge states and discrete conductance.

Conclusion

Through numerical study, the wave functions of various tight-binding lattices can be characterised. The numerical solutions of their conductance, edge states and probability density can also be easily visualised. This allows the lattices' properties to be examined theoretically, enabling possibilities of identifying properties the lattice structures might have that are similar to topological insulators.

Investigating topological insulators improves our understanding of condensed matter physics and material sciences. Being a young field, the practical applications of topological insulators might still be unclear. However, its unique properties are certain to find a use in the future. By examining the properties of various lattice structures, we hope to identify more properties similar to topological insulators that some lattice structures might have.

References

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Project Title: Characterization of Quantum Wavefunctions in a Tight-binding Lattice
Supervisor: Asst Prof Chong Yi Dong **Co-Supervisor*/Collaborator*: -**